Name: Date:

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.11.02.

1 Adjacent Angles

Two adjacent angles share a common ray. One angle measures 47° and together they form a straight angle (180°).

What is the measure of the other angle?

180° – 47° = 133°

2 Vertical Angles

Two straight lines intersect. One of the four angles formed measures 68°.

What are the measures of the other three angles?

The vertical angle = 68°. The two adjacent angles each = 180° – 68° = 112°.

3 Parallel Lines

Line AB is parallel to line CD. A transversal crosses both lines. The angle at the first intersection is 74°.

Name two other angles that also measure 74° and explain why.

The corresponding angle at the second intersection = 74°. The alternate interior angle = 74°. Both are properties of parallel lin

4 Perpendicular Lines

Lines PQ and RS are perpendicular.

What is the measure of each angle formed at the intersection? How many angles are formed?

4 angles are formed; each measures 90°.

5 Reasoning: Atlas's Claim

Atlas says that if two lines intersect and one angle is 90°, all four angles must be 90°.

Is Atlas correct? Explain using angle relationships.

Yes. If one angle is 90°, the adjacent supplementary angle = 180° – 90° = 90°, and the vertical angle = 90°. All four angles eq